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„The Golden Carrot Restaurant “

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Overview

Problem Description

The Golden Carrot Restaurant is located in Vienna with only indoor seating, and the manager of the restaurant, Maddie, who is planning to open an outdoor seating area for her guests, contacted us, the simulation experts, to conduct a simulation about her restaurant. Because summer is nearing, Maddie is planning to open an outdoor seating area for her guests. The problem is the increasing demand from guests dining outside, and this has to be adapted and adjusted to meet this demand while maintaining service quality. The current situation is that the restaurant has only an indoor seating area with limited resources.

Objectives

As for the resources, there are currently 2 waiters, 5 chefs, and 1 dishwasher with a capacity of 15 plates and a washing time of 12 minutes. There are also 15 large plates and 30 small plates.

8% of the customers - > 1 course	30,8% of the customers - > 2 courses	25,2% of the customers - > 2 courses	34% of the customers -> 3 courses
• Main	• Starter • Main	• Main • Dessert	• Starter • Main • Dessert

Abbildung 1

The restaurant offers 3 different courses start, main, and dessert, but our customers order different combinations of courses. 8% of the customers order only main, 30.8% order starter and main, 25.2% order main and dessert, and lastly, 34% order all 3 courses.

The objectives of our simulation are to expand seating capacity while ensuring that not more than 10–12% of our customers wait more than 30 minutes between ordering and receiving. This means that we have to decide which investments are necessary and optimize our resources in a cost-efficient way while maintaining high-quality service.

Input Data

Distribution

Since Maddie contacted us for our help, she also provided us with some very important information and data that could be very helpful for us to conduct the simulation properly. This data is historical data and an actual observation since Maddie observed it by herself, collected the data, and gave it to us. For our simulation, accurate input data is very important, so we are thankful to Maddie for this.

Tabelle 1

Arrival of customers per hour					
From	To	Customers wanting to sit indoors	From	To	Customers wanting to sit outdoors
11:30	12:00	11.0	11:30	12:00	3.0
12:00	13:00	28.0	12:00	13:00	4.0
13:00	14:00	38.0	13:00	14:00	8.0
14:00	15:00	25.0	14:00	15:00	12.0
15:00	16:00	xx	15:00	16:00	xx
16:00	17:00	29.0	16:00	17:00	10.0
17:00	18:00	35.0	17:00	18:00	9.0
18:00	19:00	38	18:00	19:00	12.0
19:00	20:00	32.0	19:00	20:00	8.0
20:00	20:30	12.0	20:00	20:30	2.0

Tabelle 1 shows the opening hours, time interval and customer arrival of both indoor and outdoor area. The restaurant is open between 11:30 and 21:30 but the last round of orders is taken at 20:30. Also, the type of our simulation model is terminating since it has a clear starting and ending point, which are the opening and closing hours of each day. As for customer arrivals, customers arrive per hour. We have historical data for both the indoor and outdoor seating areas. Also, the arrival of customers is given in time intervals. For example, from 14 to 15, about 25 customers arrive at the indoor seating area. But for the time interval between 15 and 16, we have missing data, so we calculated this missing data using the arithmetic mean. The number of customer arrivals between time intervals 14 and 15 is 25, and from 16 to 17 is 29, so we add these to and divide it by 2 to get the number of customer arrivals for time interval 15 to 16, which is $(25 + 29)/2 = 27$. We did the same with the outdoor seating area and got 11, which means that on average, 11 customers will arrive in the time interval of 15 to 16.

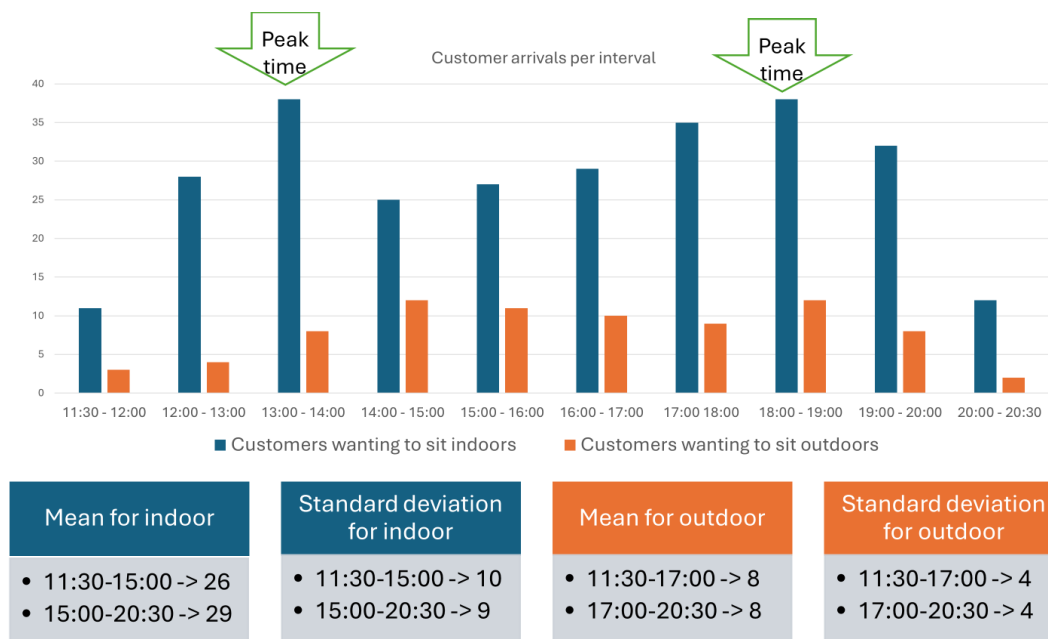


Abbildung 2

Abbildung 2 shows the normal distributions, mean and standard deviation. With the help of the customer arrival data, we obtained two normal distributions in a histogram for indoor and outdoor customer arrivals, respectively. Also, Y axis shows the number of customers, and X axis shows the time intervals. For example, in the time interval from 13 to 14, the number of outdoor customer arrivals is 38, which is also a peak time. The first normal distribution of indoor customer arrivals is between 11:30 and 15:00 and the second normal distribution of indoor customer arrivals is between 15:00 and 20:30. The first normal distribution of the outdoors is between 11:30 and 17:00 and the second normal distribution of the outdoors is between 17:00 and 20:30. For each normal distribution, we calculated the mean and standard deviation to get the average number of customer arrivals and to know how these numbers vary in each of the normal distributions, which is very important for the optimization of our resources. The highest average number of customer arrivals is 29, which is between 15 and 20:30 with a variability of 9. The average outdoor customer arrival from 11:30 to 17:00 and from 17:00 to 20:30 is 8 and the variability is 4, but when we compare outdoor with indoor, the latter varies much more, and scheduling staff members in the restaurant is very important because of these high variabilities.

Input data and blocks

- Model time in minutes
- Model simulation -> random seed
- chat -> service block, uniform (1, 2.4)
- Starter preparation -> service block, triangular (1.4, 2, 4.2)
- Main course -> service block, uniform (4.8, 7.5)
- Dessert -> service block, triangular (0.6, 1.5, 2.2)
- Eating -> service block, uniform(4, 23)
- washDishes -> service block, (12); capacity 15 plates
- washDishes -> service block, (7); capacity 50 plates.

Abbildung 3

Abbildung 3 gives an overview of the processes, time units and blocks. First, our Anylogic model's time unit is set in minutes, and the randomness of our simulation is set to a random seed.

Overview of our processes are shown in Abbildung 3. Our processes are either just in constant minutes or have uniform and triangular distributions. For instance, the preparation of the main course has a uniform distribution with a minimum and maximum value of 4.8 and 7.5. Also, we used a service block for each of these processes connected with the corresponding resource pools, such as waiters, chefs, and small and big plates. For instance, during the preparation of the main course, some of the chefs will not be available; for that reason, a service block is used.

Input data and variables

For the calculation of our costs, revenue and profit variables, event blocks and some basic codes and mathematical operations are used in our anylogic model.

Costs

Abbildung 4 shows the variables of the costs. First of all, the costs are calculated with TotalNewPlatesCost, TotalStaffCost and SplashAndCleanX2900 variables. With the help of an event

block which can be seen in Abbildung 5 the number of new small and big plates are updated. Basic setting have 30 small plates and 15 big plates. If new plates are bought then we simply subtract the original number of plates from the new number of plates and we get the number of new plates which will be used to calculate the total new plates cost with the help of the leave restaurant sink block which can be seen in Abbildung 6. Calculation of the new plates costs are done by multiplying the number of new plates with the costs of new plates which is 7.21. Lastly with the help of increasestaffcosthourly event block the total staff cost can be calculated by adding the number of staff members and then multiplying by the hourly cost of 15.63 and the total cost is also calculated which can be seen in Abbildung 7.

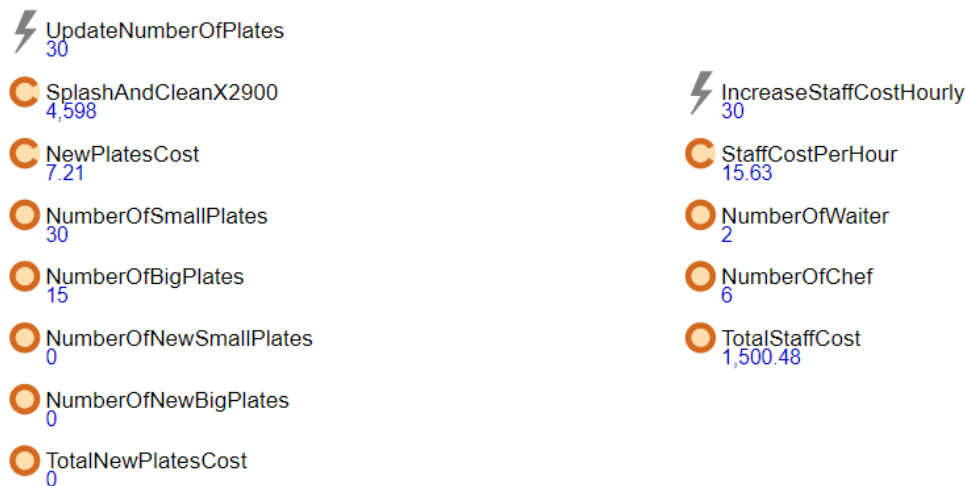


Abbildung 4

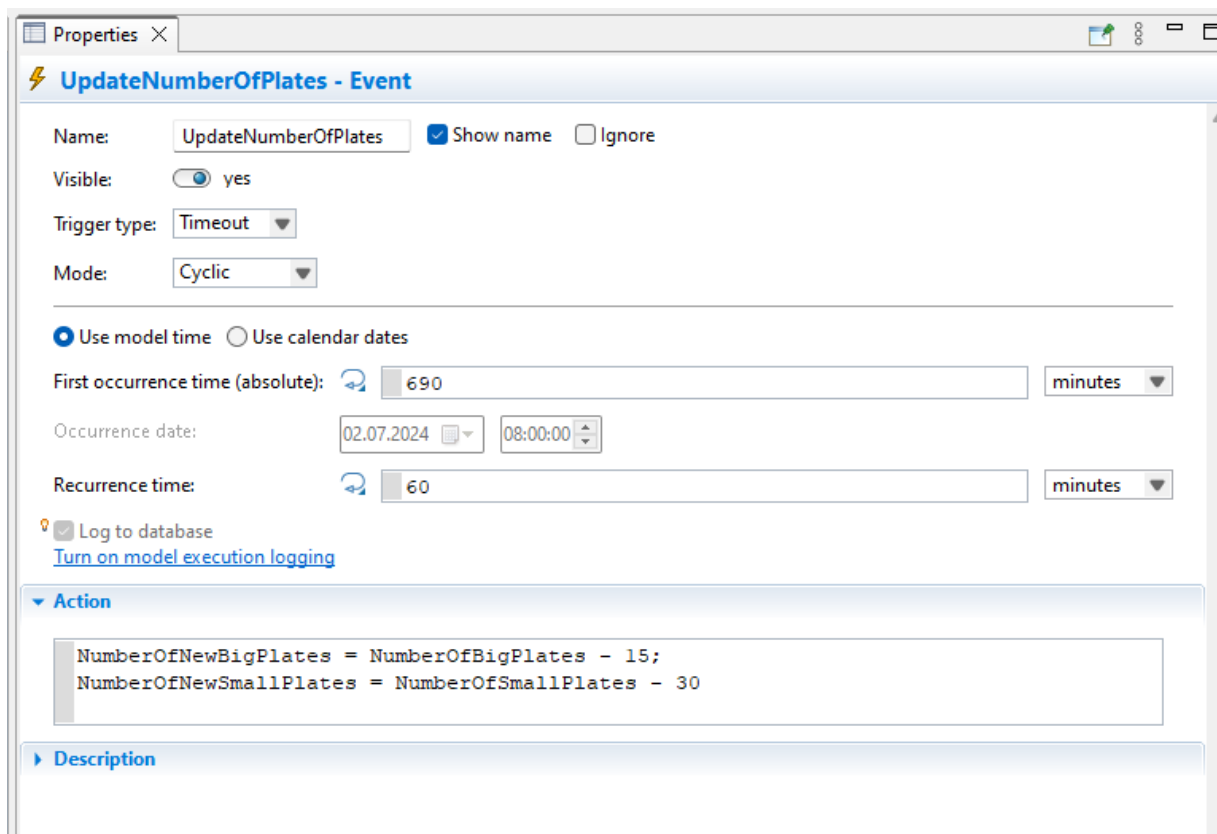


Abbildung 5

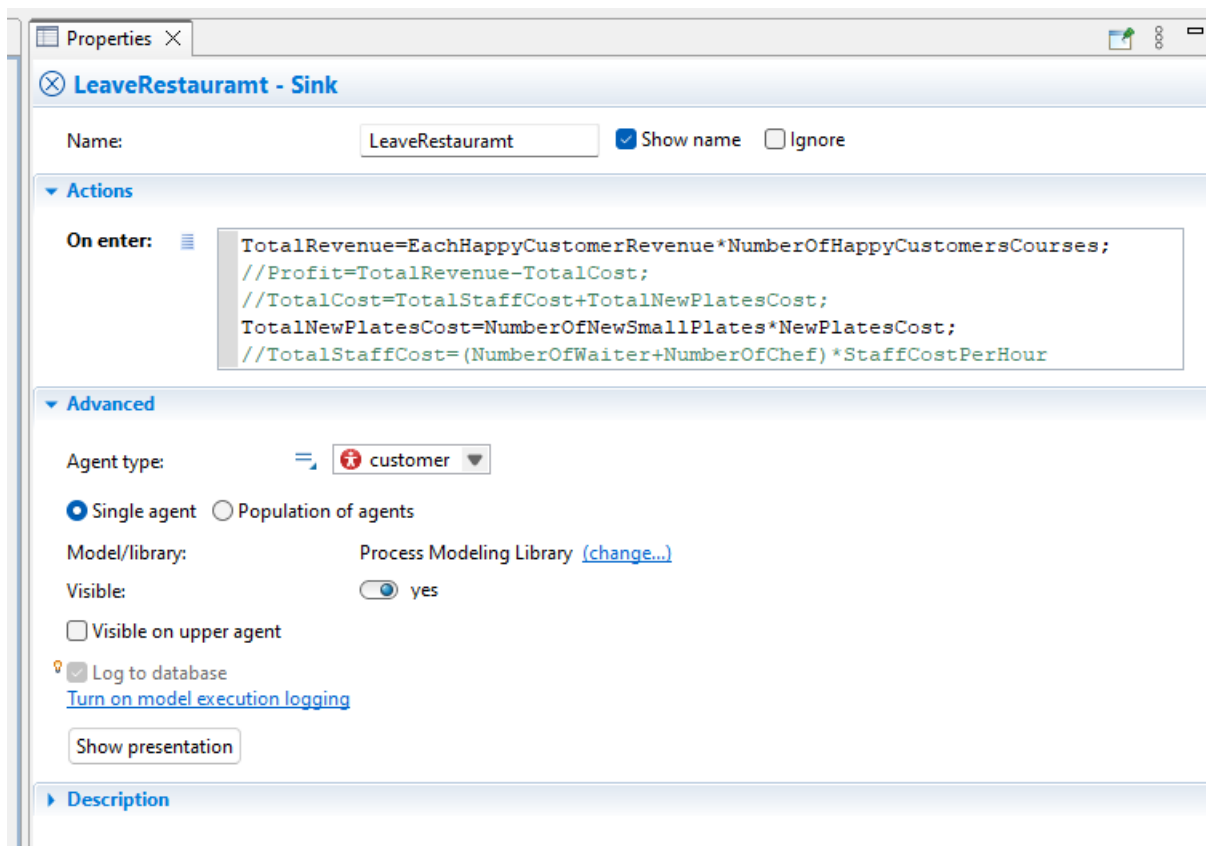


Abbildung 6

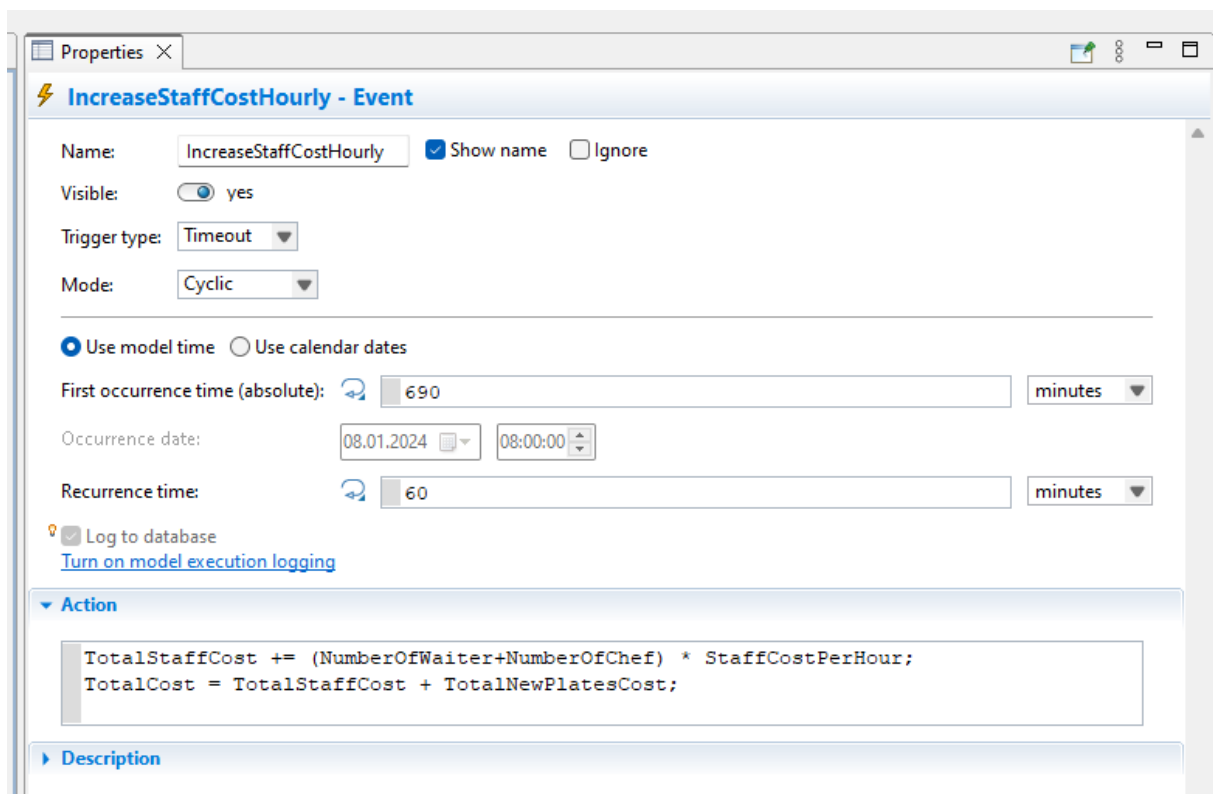


Abbildung 7

Revenue and profit

To calculate the revenue the total number of courses of happy customers are saved in the Number of happy customer courses variables and multiplied by the each happy customer revenue variable with the value of 3.88 which are shown in Abbildung 8 and 9. After calculating the revenue the profit is calculated by subtracting the totalcost from the revenue which is shown in Abbildung 9.

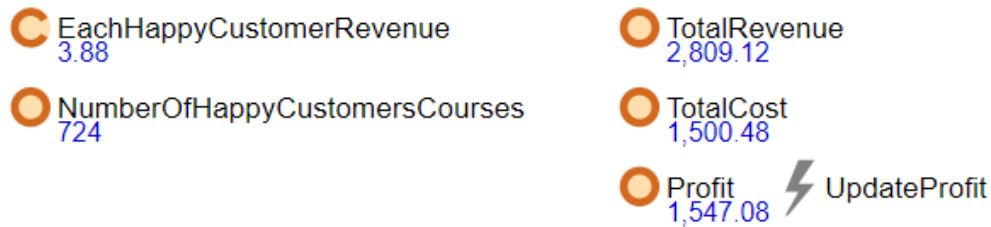


Abbildung 8

The screenshot shows the 'UpdateProfit - Event' configuration window. The 'Name' field is set to 'UpdateProfit', and the 'Visible' checkbox is checked. The 'Trigger type' is set to 'Timeout' and the 'Mode' is set to 'Occurs once'. The 'Use model time' radio button is selected. The 'Occurrence time (absolute)' field is set to '// 690' and '12 60' with a 'minutes' dropdown. The 'Occurrence date' is set to '02.07.2024' and '08:00:00'. The 'Log to database' checkbox is checked, and a link 'Turn on model execution logging' is visible. The 'Action' section contains the following code:

```
Profit = TotalRevenue-TotalCost
//Profit = TotalRevenue-TotalCost-SplashAndCleanX2900
```

The 'Description' section is currently empty.

Abbildung 9

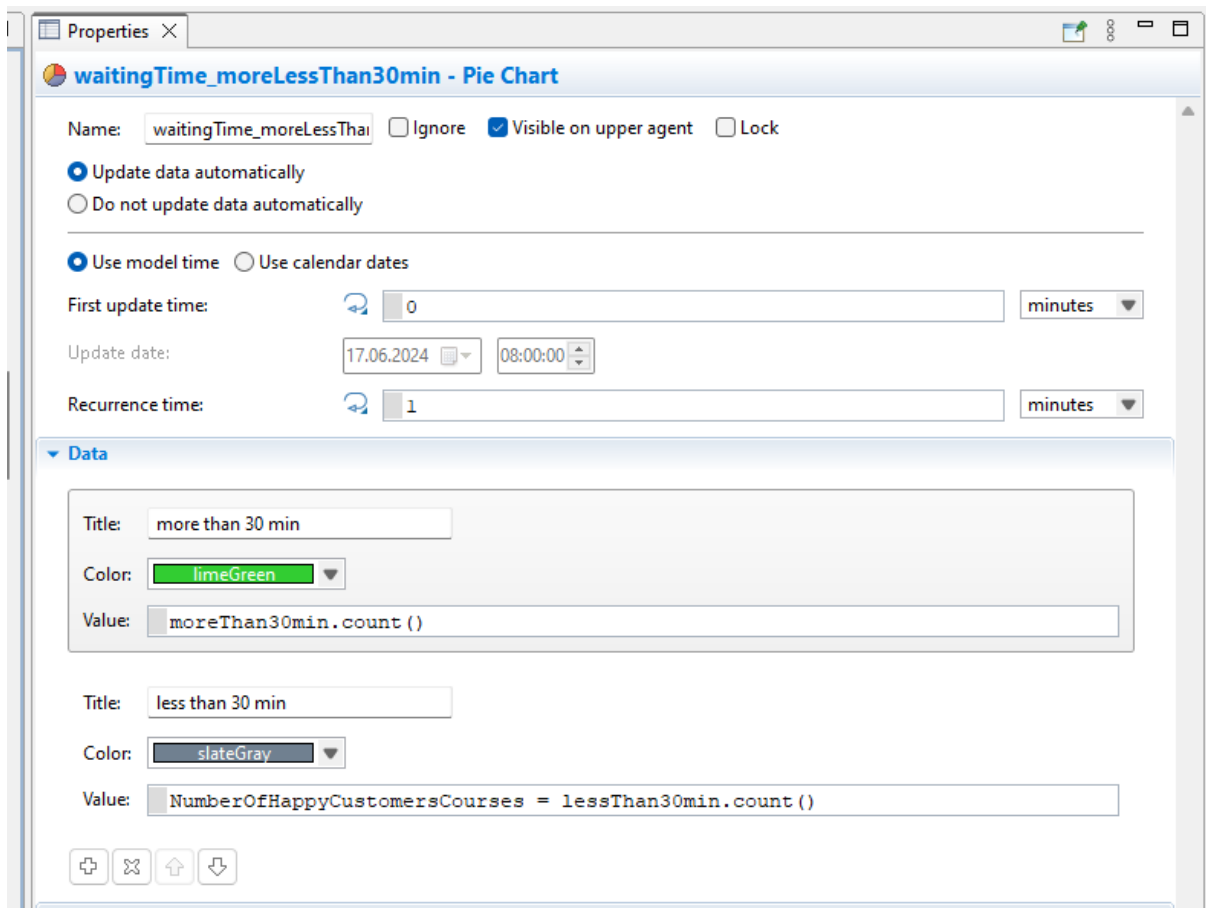


Abbildung 10

AnyLogic Model

Based on the defined distributions in the previous step, the AnyLogic model was developed and improved in an iterative process. In this chapter the final model is described, starting with a general overview followed by a deep dive into the model and its characteristics.

Structure

The model can be divided into three main components. At the source there is the arrival part (red rectangle), where the customer arrives and the corresponding courses are assigned by a function. Secondly there is the eating loop (green rectangle), where everything regarding the course preparation, serving and eating is happening, which will be repeated as long as the customer has not finished all courses he ordered initially. If he has finished, he will leave the restaurant. At last, after the customer has finished a course, the dirty dishes will be brought to the dishwasher by the waiter. The dishwashing is done in the “dishwashing-loop” (yellow rectangle) by one dishwasher and when finished the clean dishes will then be brought back to the pickup counter, so that the waiters have clean plates available.

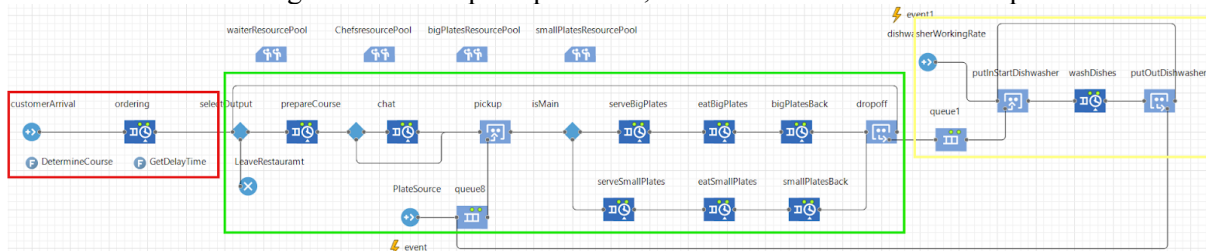


Abbildung 11

Agents

The model uses five different agents: chef, customer, dishwasher, plate and waiter. If needed the agents also contain variables to store data and keep track of different states which will be checked and changed in several points in the model.

Chef

The chef is responsible for preparing the courses the customer ordered at the beginning of the process. The total number of chefs available in the restaurant is defined in the ChefsResourcePool in the model, which also determines the utilization of all chefs. There are no further variables needed for this type of agent.

Customer

The customer is the main character in the AnyLogic model, because his order impacts all the remaining resources, like waiters, chefs, plates and the dishwasher. To keep track of the current and remaining courses, two variables are defined. First there is the integer variable “NumberOfCourses” to keep track of the number of remaining courses. The string “Course” contains the next course the customer is waiting for. Using a second string variable enables to distinguish for example between the two options for two courses, where a starter and main or a main and dessert can be ordered.

At last there is a double variable “time” to measure the waiting time for each course.

Dishwasher

The dishwasher is used to clean the dirty dishes and has a capacity constraint. To implement the capacity restriction in the model, two variables are assigned. The integer variable “Capacity” is a static variable which is set initially and will not be changed during runtime. The integer variable “ItemsLoaded” is set to 0 at the beginning and keeps track of the plates which are carried by the dishwasher. This variable gets increased everytime a plate is added to the dishwasher and will be reset to 0 if the dishes are dropped at the end.

Plate

The plates are required to carry the courses after they were prepared by the chefs. The amount of big and small plates are defined in the corresponding Resource Pools “bigPlatesResourcePool” and “smallPlatesResourcePool” in the model. There are no variables needed for this agent type.

Waiter

The waiters are responsible for taking orders, serving courses and bringing back the dirty plates to the kitchen. The number of waiters available is defined in the Resource Pool “waiterResourcePool”. To maintain the utilization, the waiters are assigned to the corresponding service blocks as a resource set. Hence there are no variables needed.

Main Components

In this section we go more into detail and take a closer look at the three main components, with a focus on the implementation and the underlying data structure of the model.

Arrivals

In order to assign the different orders according to the given distributions to each customer entering the system, a function called “DetermineCourse” is used at the source “customerArrival” of the model.

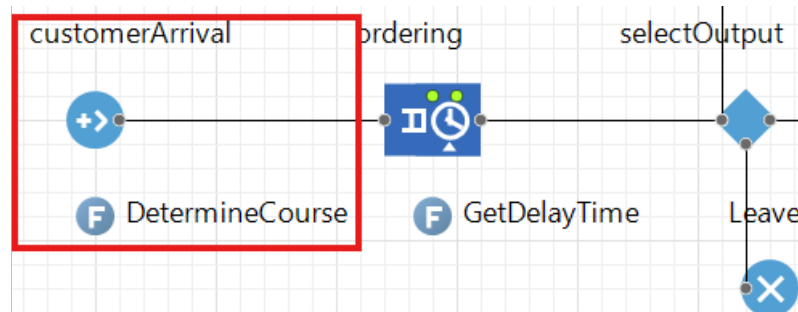


Abbildung 12

The function “DetermineCourse” is a void function, which means it does not return a value, because it initializes the two variables “NumberOfCourses” and “Course” of the agent customer only. To do so, it is required to pass the customer as an argument.

In the function a random variable between 0 and 1 is generated, to assign an order according to the probabilities given in the task. For example if the customer orders only one course, the variable “NumberOfCourses” is initialized with 1 and “Course” is set to “Main”. These variables are crucial for the whole model and maintain the correct state of each customer.

▼ Function body

```
double randomNumber = uniform();
if(randomNumber < 0.1){
myAgent.NumberOfCourses = 1;
myAgent.Course = "Main";
} else if(randomNumber < 0.352){
myAgent.NumberOfCourses = 2;
myAgent.Course = "Main";
} else if(randomNumber < 0.66){
myAgent.NumberOfCourses = 2;
myAgent.Course = "Starter";
} else{
myAgent.NumberOfCourses = 3;
myAgent.Course = "Starter";
}
```

Abbildung 13

The function “DetermineCourse” is called in the source “CustomerArrival” at action “On exit” with agent as an argument.

▼ Actions

On before arrival:	
On at exit:	
On exit:	DetermineCourse (agent)

Abbildung 14

After the arrival of the customer, there is a Service Block called “ordering” to implement the time which is needed to place an order by the customer. As there is also a waiter required for this step, the Resource Pool “waiterResourcePool” is assigned to the Resource set by using one waiter.

Resource sets (alternatives):

waiterResourcePool	1
--------------------	---

+ Add list

Abbildung 15

Eating-Loop

The eating-loop contains all steps and procedures starting from the preparation of each course, followed by picking up and serving the courses, eating and bringing back the dirty dishes. To ensure a proper use of each resource like plates, waiters and chefs, the corresponding Resource Pools are assigned to the Service Blocks.

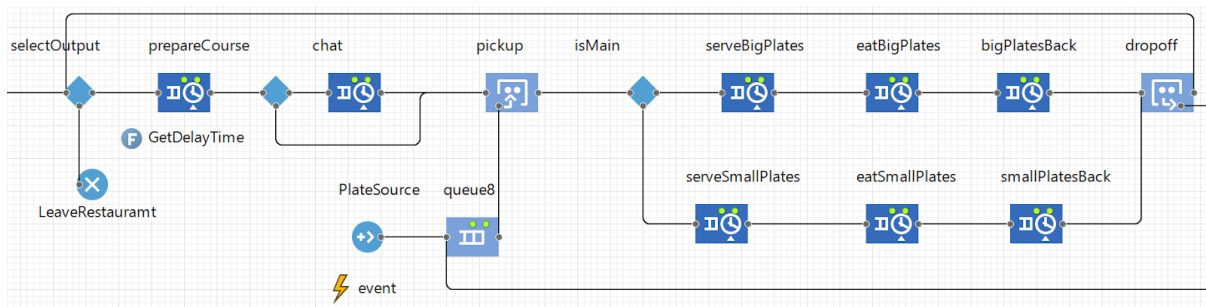


Abbildung 16

At the beginning it is necessary to determine which course needs to be prepared by the chefs in the Service Block “prepareCourse”. This is done by using a function “GetDelayTime” which uses the customer as an argument. Based on the variable “Course” the corresponding delay time will be returned by the function.

Function body

```
double delayTime = 0;
if(agent.Course == "Starter"){
    delayTime = triangular( 1.4, 4.2, 2.0);
} else if(agent.Course == "Main"){
    delayTime = uniform(4.8,7.5);
} else if (agent.Course == "Dessert"){
    delayTime = triangular( 0.6, 2.2,1.5 );
}
return delayTime;
```

Abbildung 17

The Service Block “prepareCourse” uses one chef from the “chefsResourcePool” and the function “GetDelayTime(agent)” to determine the delay time.

Resource sets (alternatives): 1

Maximum queue capacity:

Delay time: minutes

Abbildung 18

After the course is prepared, it can be picked up by the waiter. As stated in the task, the waiters chat 15% of the time with chefs when picking up the meal. To realize that, a select output element is used to send the waiter with a probability of 0.15 to a chat Service Block, where the “chefsResourcePool” and “waiterResourcePool” are assigned as resource sets.

Resource sets (alternatives): 1

Maximum queue capacity:

Delay time: minutes

Abbildung 19

Then the course can be picked up by the waiter. To do so, the corresponding plates are required. In the initial setup, there are 15 big plates and 30 small plates available. To realize that, at the start of the simulation, 45 plates are injected at the source “PlateSource” in the system by using an event.

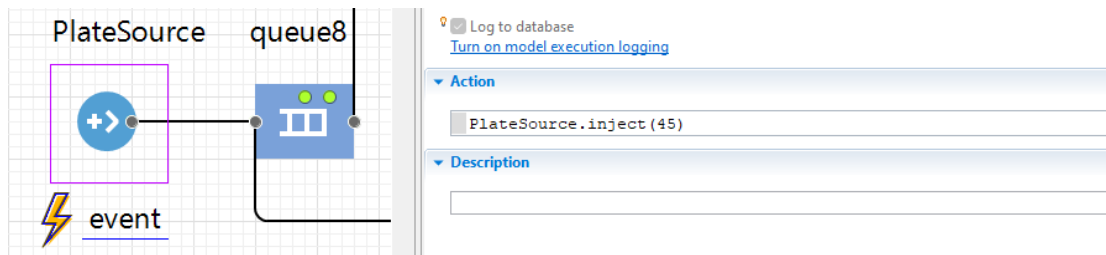


Abbildung 20

Those plates are then available at the pickup queue. At the pickup element, for each course one plate is picked up. Afterwards a select output element is used to determine if the course is a main or not. This is required to distinguish between the two different plate types. In the upper path, the main is carried, where big plates are assigned to the resource sets and in the lower path the small plates for starters and desserts are used. Furthermore, the “waiterResourcePool” is assigned to the “serveBigPlates”, “serveSmallPlates”, “bigPlatesBack” and “smallPlatesBack” service blocks. After the waiters brought back the plates, the customer and the plate is splitted at the dropoff element. The customer goes back to the start of the loop and the plate is forwarded to the dishwasher loop.

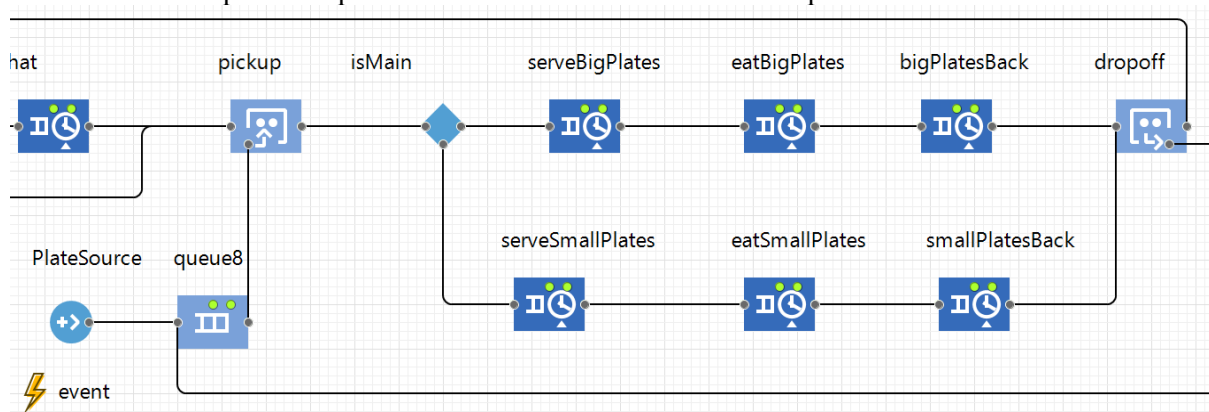


Abbildung 21

If the customer has finished the current course, the two variables “NumberOfCourses” and “Course” are maintained in the service block “eatBigPlates” or “eatSmallPlates” depending on the current course. In both scenarios, the variable indicating the remaining number of courses “NumberOfCourses” is decreased by one and the next course is set depending on the current course. If the customer has finished his last course, the variable “Course” will be set to “Exit”.

On exit:

```

if(agent.NumberOfCourses == 3){
  agent.NumberOfCourses--;
  agent.Course = "Main";
}else if(agent.NumberOfCourses == 2 && agent.Course == "Starter"){
  agent.NumberOfCourses--;
  agent.Course = "Main";
}else if(agent.NumberOfCourses == 2 && agent.Course == "Main"){
  agent.NumberOfCourses--;
  agent.Course = "Dessert";
}else{
  agent.NumberOfCourses--;
  agent.Course = "Exit";
}

```

Abbildung 22

When the customer comes back to the start of the eating-loop, the select output checks if the customer is waiting for his next course or if he has already finished and leaves the restaurant. As long as the variable “Course” does not reach the state “Exit”, the eating-loop will be repeated.

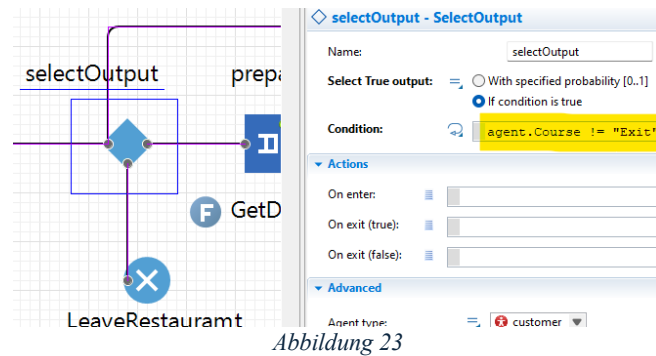


Abbildung 23

Dishwasher-Loop

At the beginning of the simulation, one dishwasher is injected to the dishwasher-loop by using an event.

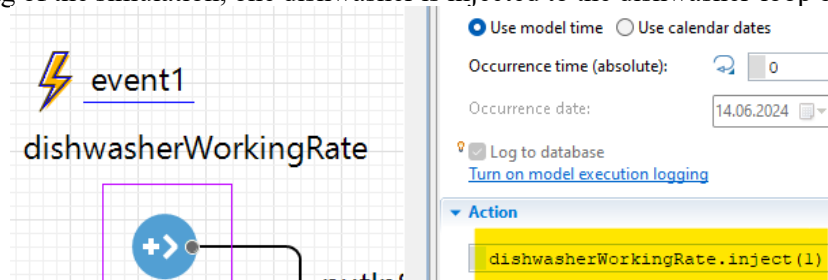


Abbildung 24

The dishwasher then picks up plates that are waiting in the queue before the pickup element. To ensure that the dishwasher does not exceed its capacity, a pickup condition is defined which says that the “ItemsLoaded” cannot exceed the “Capacity” variable. The variable “ItemsLoaded” is increased by one with every plate that will be picked up.

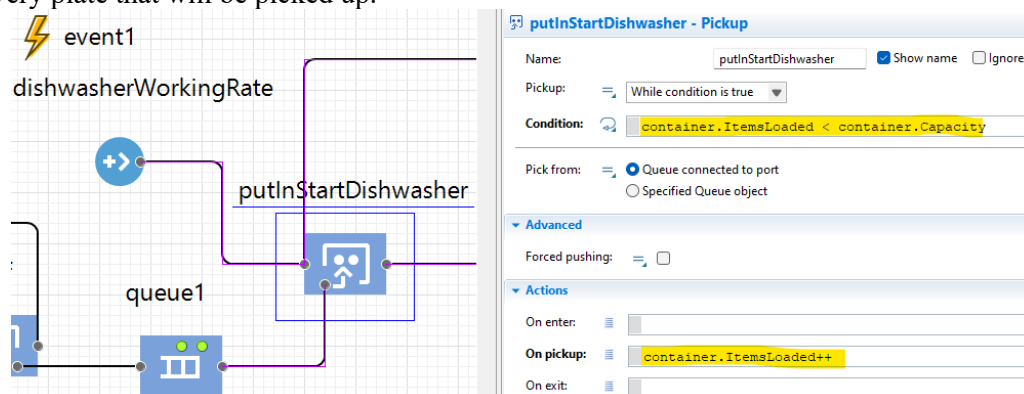


Abbildung 25

After all plates that fit in the dishwasher have been picked up, the cleaning procedure “wahsDishes” can start and takes 12 minutes. When the dishwasher has finished cleaning the plates, they will get back to the pickup waiting queue and the dishwasher will go back to the start of the dishwasher-loop to pick up the next round of dirty dishes. At the drop off element, the variable “ItemsLoaded” will be reset to 0.

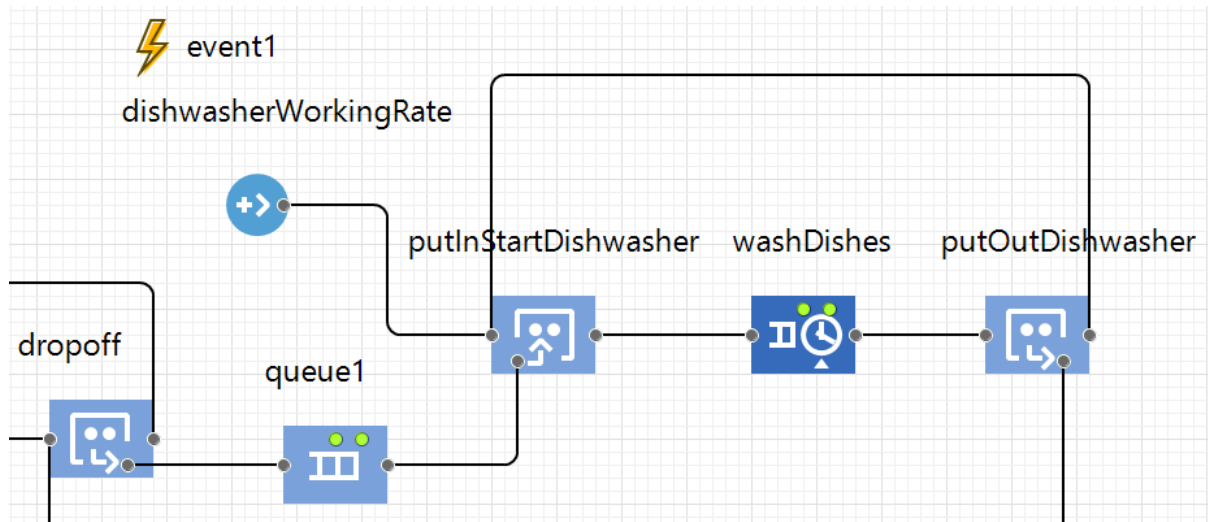


Abbildung 26

Entity Flow

The entity flow in the AnyLogic model consists of three agent types. The central aspect of the simulation is the customer agent, which starts at the source and walks through the eating-loop. Then there are the plates which start at the pick up element in the eating-loop and then go to the dishwasher loop, where the plates are cleaned and get back to the pickup queue. At last there is the dishwasher, which stays in the dishwasher loop. The other agents are managed by resource pools and are assigned to the corresponding service blocks where they are needed.

To illustrate the entity flow in the model, the flow of each agent is shown in the figure below. The red lines represent the customers, green lines the plates and orange lines are for the dishwasher.

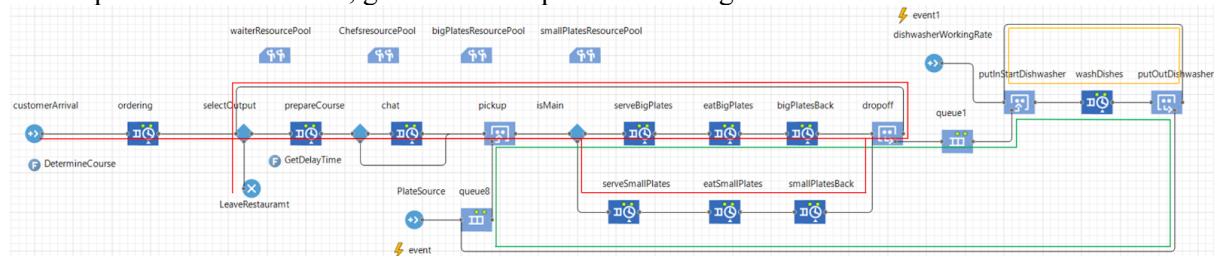


Abbildung 27

Experimental Design

The AnyLogic model described in the previous section was used to test different scenarios. These scenarios were created by setting different configurations of the models parameters. This section will give an overview of the parameters, manipulated to create these scenarios, the different tested scenarios, the configurations of the parameters and will briefly discuss the number of replications for the different scenarios.

Parameters

To create different scenarios in our AnyLogic model, some Parameters had to be manipulated. This section will outline these parameters.

Seating area

By opening the outdoor seating area Maddie expects an increase of customer arrivals per hour (section 2.1.). To implement this, we created two different Schedules in AnyLogic,

“customerArrivalScheduleInside” and “customerArrivalScheduleInsideOutside”. We used these Schedules as Rate schedules for the “customerArrival” - Source.

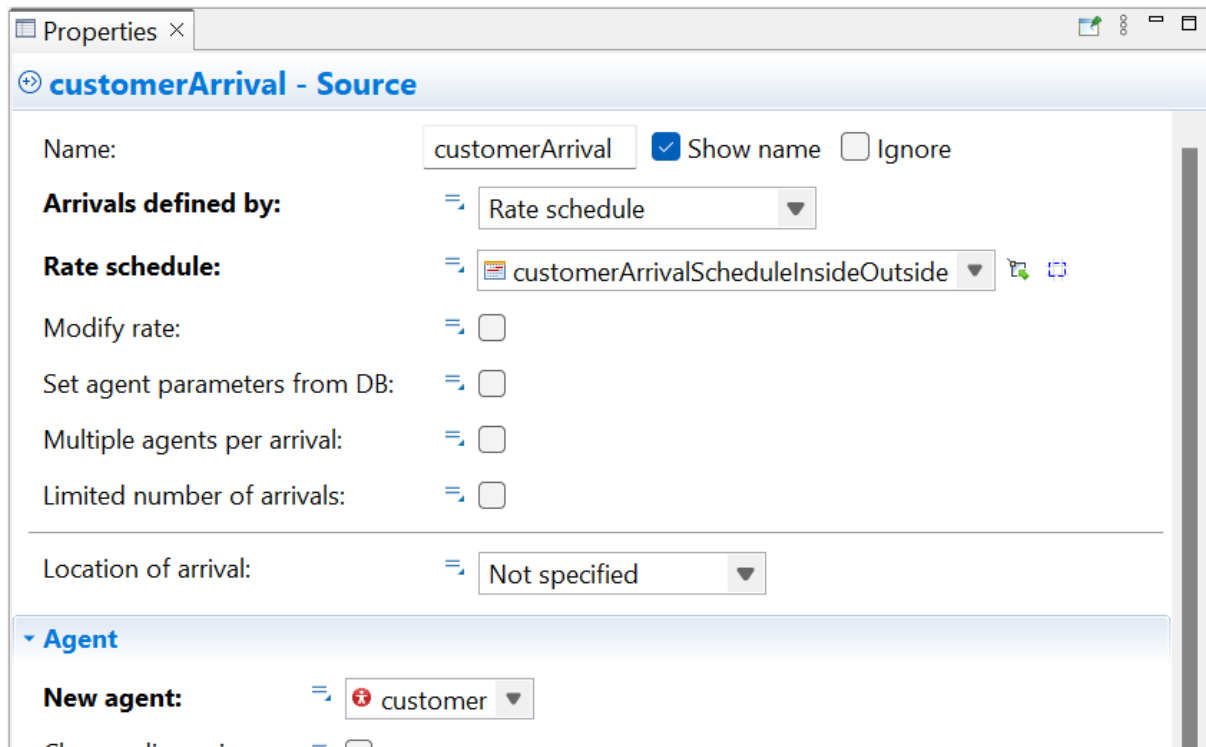


Abbildung 28

Dishwasher

As mentioned before, we created an Agent dishwasher. This dishwasher-agent holds two parameters “Capacity” and “ItemsLoaded”. In the dishwasher-loop, described in section 3.3.3., we use a service block “washDishes” to simulate the dishwasher running time.

The current dishwasher and the potential new one (Splash And Clean X2900), differ in two aspects. While the current one has a capacity of 15 plates and a runtime of 12 minutes, the “Splash And Clean X2900” can hold up to 50 plates and has a runtime of only 7 minutes.

To set the dishwasher parameter, we need to manipulate the capacity-parameter of our agent (from 15 to 50) and our “washDishes” service block (Delay Time 12 to 7 minutes).

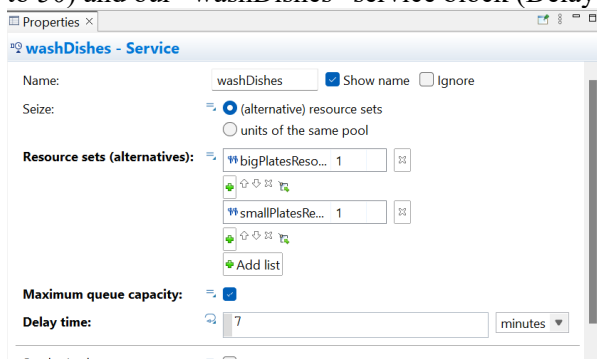


Abbildung 29

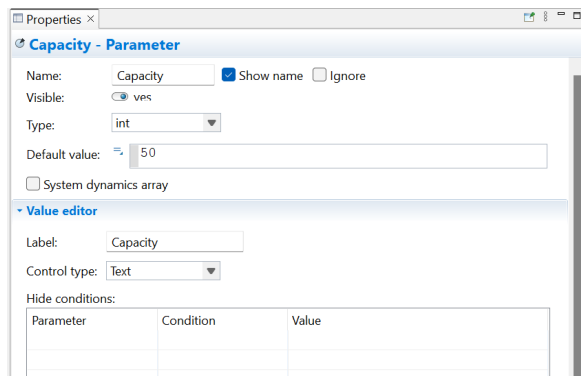


Abbildung 30

Waiters

The waiters are represented by a resource pool, in which the capacity is connected to a variable “NumberOfWaiters”.

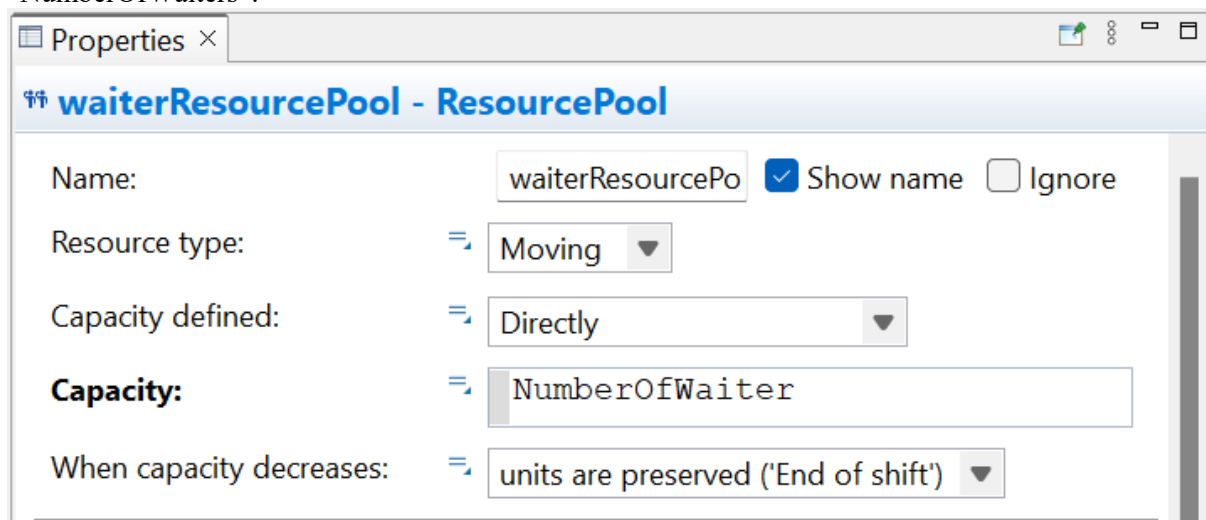


Abbildung 31

To change the number of waiters we have to manipulate the “NumberOfWaiters” variables initial value (e.g. from 2 to 3).

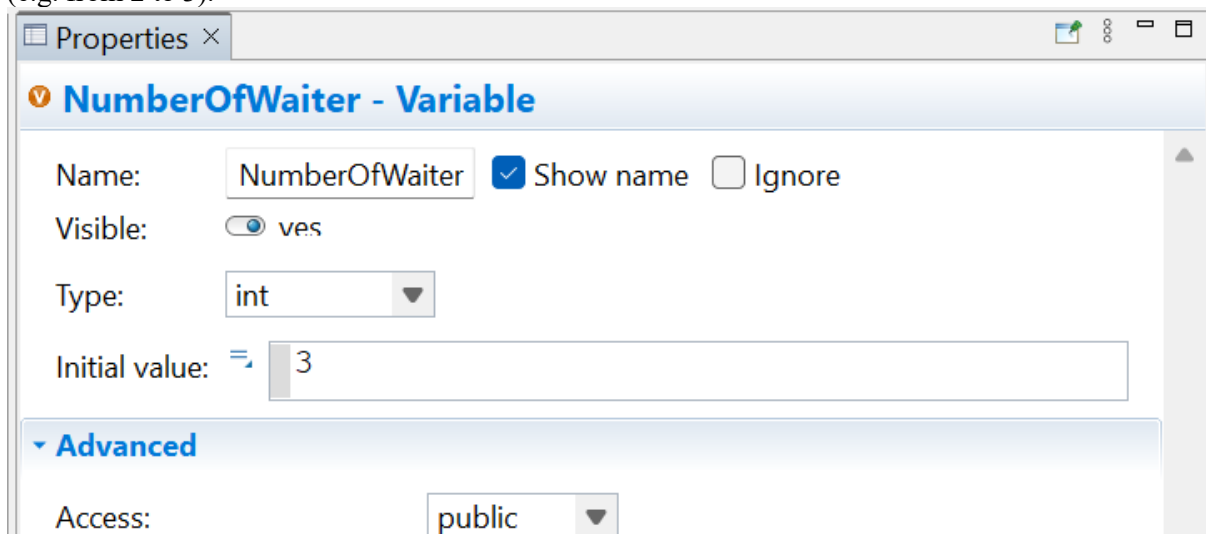


Abbildung 32

Chefs

Like the waiters, the chefs get represented by a resource pool connected to a variable. Therefore the chefs parameter can be changed just like the waiters parameter.

Plates

As there are two different types of plates for starters and desserts (small plates) and for main course (big plates) plates display two types of parameters, which must be distinguished. For each of these parameters “small plates” and “big plates” we created a resource pool and a variable, connected the same way as for waiters and chefs. These parameters can be changed the same way.

Scenarios

The different scenarios were set in the AnyLogic model by manipulating the parameters, introduced in the previous section. They got tested on the main objective (not more than 10-12% of customers have to wait longer than 30 minutes for their next course) and evaluated on the expected profit. Before we lay our focus on interpreting the results in section 5., this section will deal with the investigated scenarios and if they meet the main objective. As the number of tested scenarios would go beyond the scope of this report only the main steps to finding the optimal solution will be outlined.

Initial situation (scenario 1)

In the initial situation, in which Maddie calls us, she employs two waiters and five cooks. In her restaurant she has 30 small and 15 big plates and the old dishwasher. Only the inside seating area is open.

Tabelle 2

Parameter	manipulation	value
seating area	Rate Schedule	customerArrivalScheduleInside
dishwasher	Capacity parameter	15
dishwasher	washDishes service	12
waiters	NumberOfWaiter variable	2
chefs	NumberOfChef variable	5
small / big plates	small / big plates variable	30 / 15

We let this configuration run 4 times.

Tabelle 3

number of runs	4
mean % time under 30 minutes	2,70 %
mean profit	214.351,52€

Opening the outdoor seating area (scenario 2)

Now we want to check if we still meet the main objective, when opening the outdoor seating area.

Tabelle 4

Parameter	manipulation	value
seating area	Rate Schedule	customerArrivalScheduleInsideOutside
dishwasher	Capacity parameter	15
dishwasher	washDishes service	12
waiters	NumberOfWaiter variable	2
chefs	NumberOfChef variable	5
small / big plates	small / big plates variable	30 / 15

We only let this one run 3 times, because it didn't look too promising.

Tabelle 5

number of runs	3
mean % time under 30 minutes	58,26 %
mean profit	20.043,35€

The main objective is missed by far so we had to adjust some other parameters.

Buying new dishwasher (scenario 3)

Tabelle 6

Parameter	manipulation	value
seating area	Rate Schedule	customerArrivalScheduleInsideOutside
dishwasher	Capacity parameter	50
dishwasher	washDishes service	7
waiters	NumberOfWaiter variable	2

chefs	NumberOfChef variable	5
small / big plates	small / big plates variable	30 / 15

Tabelle 7

number of runs	4
mean % time under 30 minutes	43,95 %
mean profit	77.171,37€

Buying more plates (scenario 4)

As one time investments, like a dishwasher and plates are less expensive than additional employees, we tried out these configurations first.

Tabelle 8

Parameter	manipulation	value
seating area	Rate Schedule	customerArrivalScheduleInsideOutside
dishwasher	Capacity parameter	50
dishwasher	washDishes service	7
waiters	NumberOfWaiter variable	2
chefs	NumberOfChef variable	5
small / big plates	small / big plates variable	50 / 35

As in the scenario before we quickly saw that the main objective couldn't be met, so we let this scenario run two times. We saw that one time investments wouldn't help to maintain service quality when opening the outdoor seating area.

Tabelle 9

number of runs	2
mean % time under 30 minutes	46,49 %
mean profit	66.945,83€

Employing a new waiter (scenario 5)

Tabelle 10

Parameter	manipulation	value
seating area	Rate Schedule	customerArrivalScheduleInsideOutside
dishwasher	Capacity parameter	50
dishwasher	washDishes service	7
waiters	NumberOfWaiter variable	3
chefs	NumberOfChef variable	5
small / big plates	small / big plates variable	50 / 35

number of runs	2
mean % time under 30 minutes	41,77 %
mean profit	62.028,04€

Employing a new chef (scenario 6)

Tabelle 11

Parameter	manipulation	value
seating area	Rate Schedule	customerArrivalScheduleInsideOutside
dishwasher	Capacity parameter	50
dishwasher	washDishes service	7
waiters	NumberOfWaiter variable	2
chefs	NumberOfChef variable	6
small / big plates	small / big plates variable	50 / 35

As the increased number of waiters wasn't enough to meet the objective, we increased the number of chefs. We let this configuration run 4 times and identified the number of chefs as the bottleneck for obtaining service quality when opening the outdoor seating area.

Tabelle 12

number of runs	4
mean % time under 30 minutes	3,30 %
mean profit	230.731,34€

Results

The following section will first give a summary of the results before these results are discussed.

Summary of Results

After finding out that the most important component for this simulation was the number of chefs, we tried some configurations, including not buying a new dishwasher, but only investing in new plates and others. The results (% of waiting time longer than 30 minutes and estimated revenue) are shown in the following figure.

(For Settings: inside/outside_old/newdishwasher_#waiter_#chefs_#smallplates_#bigplates)

Tabelle 13

Scenario	Setting	% > 30 min	profit
1	i_o_2_5_30_15	2,70%	214.351,52 €
3	o_n_2_5_30_15	43,95%	77.171,37 €
4	o_n_2_5_50_35	46,49%	66.945,83 €
-	o_n_2_6_30_15	3,85%	227.851,10 €
-	o_n_2_6_40_25	3,68%	228.966,51 €
6	o_n_2_6_50_35	3,30%	230.731,34 €
optimal	o_n_2_6_60_45	3,18%	230.937,70 €
-	o_n_2_6_70_55	3,16%	230.608,81 €
-	o_n_2_6_80_65	3,03%	230.882,65 €
5	o_n_3_5_50_35	41,77%	62.028,04 €
-	o_n_3_6_50_35	3,75%	206.896,95 €
2	o_o_2_5_30_15	58,26%	20.043,35 €
-	o_o_2_6_400_350	2,67%	226.365,17 €

This figure only shows the mean of the investigated objectives for the different scenarios. An Excel file containing information about investment costs (dishwasher, plates, employees), the expected revenue and the expected profit is appended (“ErgebnisseReport.xlsx”).

Discussion of Results

As mentioned before, the one time investments (dishwasher and plates) don’t have a big influence on the performance but also on the profit. Over a timespan of around 50.000 customers (maximal number for AnyLogic base model agents) these investments are amortized rather quickly.

Employing an additional waiter isn’t enough as well. Without increasing the number of chefs we can never meet the objective, that not more than 10-12% of customers have to wait longer than 30 minutes for their next course.

The optimal scenario we found was to buy the new dishwasher, 30 additional plates small and big each and employ an additional waiter. This way the service quality can be obtained when opening the outdoor seating area, while the profit over the 50.000 customers gets maximized. For this scenario the break even point is reached only after approximately 3 days.

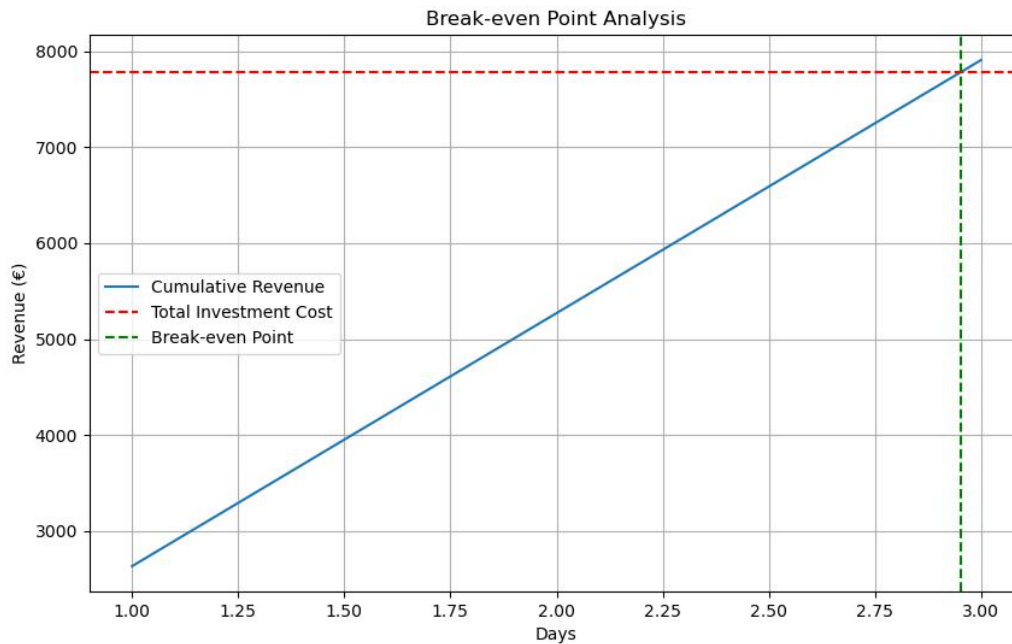


Abbildung 33

For this break even analysis we used python code (executed in “JupyterNotebook”). The code (“breakeven.ipynb”) as well as a csv-file containing the calculations made (“break_even_analysis.csv”) are appended.

Conclusions and recommendations

As the scenarios and results, presented in the previous sections show, failing to obtain service quality will lead to a decrease in profit. To meet the objective and obtain service quality it is crucial to employ an additional chef. Based on these results we would recommend to open the outdoor seating area, buy the new dishwasher and 30 new plates of each size and employ an additional chef. Through this Maddie can obtain service quality in her restaurant and maximizes her profits. Compared to the initial situation in which she didn’t open the outdoor seating area and no additional investments were made, her estimated profit will rise by roughly 6.000€ over a period of 50.000 customers.